7.1 Inverse Variation

Name _____ Period ____ Date _____

Algebra II 7.1 · Student sheet

I can…

- tell whether x and y show direct variation, inverse variation, or neither.
- write and use inverse variation equations.
- use inverse variation to solve real-life problems.

Vocabulary · fill in the missing word

Direct variation — $y = ax$, $a \neq 0$: $\frac{y}{x}$ is _____

Inverse variation — $y = \frac{a}{x}$, $a \neq 0$: the product xy is _____

Constant of variation — _____

Part 1 · Pizza party

A party has 24 slices of pizza, shared equally. Fill in the table.

People (x)	1	2	3	4	6	8	12
Slices each (y)							
x · y							

Plot the points on graph paper. What stays the same? Write an equation for y: _____

Part 2 · Notes and examples (fill in each note, then work the example)

Direct variation: y = _____, so y/x is _____.

Inverse variation: y = _____, so the _____ xy is constant.

Example 1

Tell whether x and y show direct variation, inverse variation, or neither.

a. $xy = 5$ b. $y = x - 4$ c. $3y = 2x$ d. $x = \frac{7}{y}$

Example 2

Tell whether x and y show direct variation, inverse variation, or neither.

x	2	4	6	8
y	18	9	6	4.5

Writing an inverse variation: To write an inverse variation equation, find a = _____, then write y = a/x.

Example 3

x and y vary inversely, and y = 6 when x = 4. Write an equation that relates x and y. Then find y when x = 3.

Example 4

The time t (in hours) it takes to drive a route varies inversely with your speed r (in miles per hour). At 60 miles per hour, the drive takes 3 hours. How long does it take at 45 miles per hour?

Part 3 · Partner practice

Partner 1

x and y vary inversely, and y = −5 when x = 3. Find y when x = −6.

Partner 2

Tell whether x and y show direct variation, inverse variation, or neither. Write an equation if possible.

x	3	6	9	12
y	12	6	4	3